by default, allowing direct execution of supported TypeScript syntax without additional flags.

Future versions of Node.js aim to provide full native support for TypeScript, eliminating the need for any flags.

The importance of this development cannot be overstated. Native TypeScript support in Node.js eliminates the need for additional build steps and tools, streamlining the development process and improving productivity. As Node.js continues to evolve, full native support for TypeScript might become a game-changer for JavaScript developers, enabling them to write more maintainable, efficient, and scalable code.

As TypeScript continues to grow in popularity, new developments are emerging in the JavaScript ecosystem. Microsoft has proposed a Stage 0 feature for JavaScript that introduces optional and erasable type annotations, inspired by TypeScript. These annotations can be used by type checkers like TypeScript but are ignored by JavaScript engines, effectively treating them as comments. This proposal aims to bridge the gap between JavaScript and TypeScript, thereby making it easier for developers to adopt typed programming practices without requiring full TypeScript adoption.

For example, with this proposal, JavaScript code could look like what is shown in listing 3.23. Type checkers (e.g., TypeScript) can validate the types, and JavaScript engines will treat the type annotations as comments and ignore them during execution.

Listing 3.23 Proposed JavaScript code for implementation

```
function add(a: number, b: number): number {
  return a + b;
}
```

The rise of evergreen browsers (browsers that automatically update to the latest version) further reduces the need for transpilation. Since most users now run up-to-date browsers with support for modern JavaScript features, developers can rely on native browser capabilities without worrying about backward compatibility. Combined with the optional type syntax proposal,